

Phase 2 – Scanning

Objective:

The objective of the scanning phase is to identify open ports, active network services, service versions, and operating system information running on the target machine. The information gathered during this phase helps identify potential attack surfaces and vulnerabilities that may be exploited in later stages of the penetration testing process.

Purpose:

Scanning is the second phase of penetration testing. After verifying that the target machine is reachable during the reconnaissance phase, the penetration tester performs network scanning to discover accessible services and collect technical information about the target system.

Target Machine: Metasploitable2

Target IP Address: 192.168.56.101

Scanning Tool: Nmap

Attacker Machine: Kali Linux

Step 1: Perform a Basic Port Scan:

A basic TCP port scan was performed using Nmap to identify open ports on the target machine.

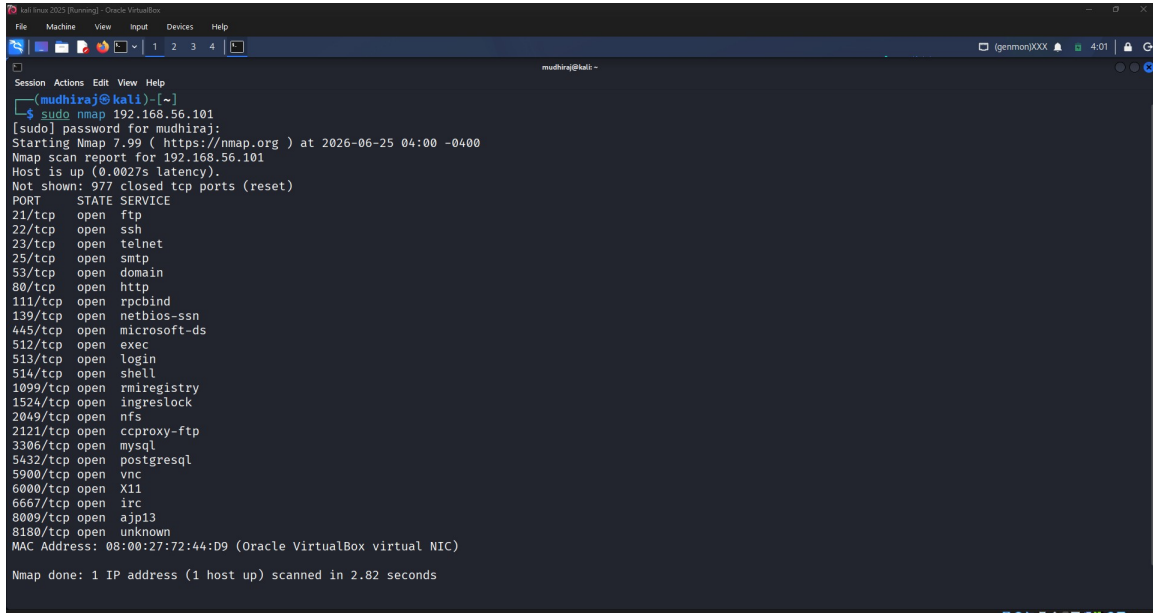
Command Used :

```
sudo nmap 192.168.56.101
```

Command Explanation:

- `nmap` – Launches the Network Mapper tool.
- `192.168.56.101` – Specifies the IP address of the target machine to be scanned.

Screenshot:



```
mudhiraj@kali:~$ sudo nmap 192.168.56.101
[sudo] password for mudhiraj:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-25 04:00 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:72:44:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.82 seconds
```

Figure 3: Basic Nmap port scan showing open TCP ports on the Metasploitable2 target machine.

Observation:

The basic scan successfully identified several open TCP ports on the target machine. These open ports indicate active network services that may contain known vulnerabilities and require further investigation.

Step 2: Detect Service Versions:

After identifying the open ports, a service version detection scan was performed to determine the exact software versions running on each service.

Command Used:

```
sudo nmap -sV 192.168.56.101
```

Command Explanation:

- -sV – Enables service version detection.
- 192.168.56.101 – Target machine IP address.

Screenshot

```
mudhiraj@kali:~$ sudo nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-25 04:01 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00070s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rsh rexecd
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-Subuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:72:44:D9 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.26 seconds
```

Figure 4: Nmap service version detection (-sV) identifying software versions running on the target machine.

Observation:

The scan successfully identified the versions of the running services. The detected software versions can be compared with publicly available vulnerability databases to determine whether known security vulnerabilities exist.

Step 3: Perform an Aggressive Scan:

An aggressive scan was executed to collect additional information about the operating system, service versions, default NSE script results, and network path.

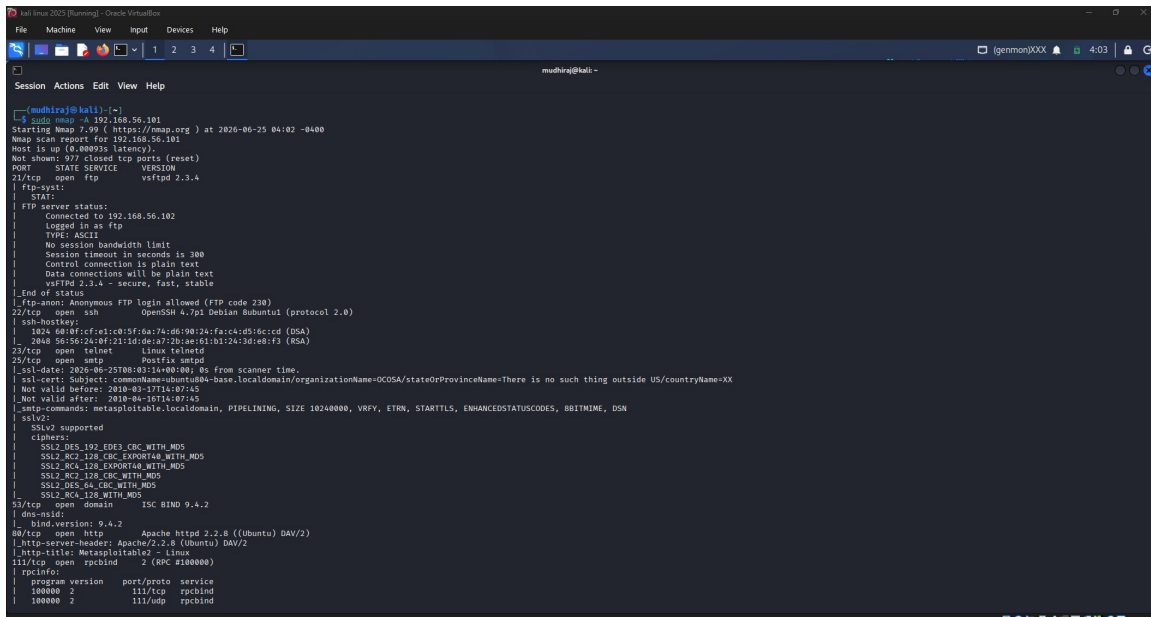
Command Used:

```
sudo nmap -A 192.168.56.101
```

Command Explanation:

- -A – Enables aggressive scanning, including:
 - Operating System Detection
 - Version Detection
 - Default NSE Scripts
 - Traceroute

Screenshot:



```
mudhir@kali:~$ sudo nmap -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2020-06-25 04:02 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00092s latency).
Not shown: 927 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-sys:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.56.102
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
|_ssh-hostkey:
|_  1024 60:8f:cf:a1:c0:5f:6a:74:d6:90:24:fa:c4:05:8c:cd (DSA)
|_  2048 56:56:24:8f:21:1d:de:a7:2d:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet  Linux telnetd
25/tcp    open  smtp     Postfix smtpd
|_ssl-date: 2020-06-25T00:03:16+00:00 0s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-06-10T14:07:45
|_smtp-command: setheader,localdomain, PIPELINING, SIZE 1024000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslsz:
|_  SSLv2 supported
|_  ciphers:
|_    SSLv2_DES_192_EDE3_CBC_WITH_MD5
|_    SSLv2_RC2_128_CBC_EXPORT40_WITH_MD5
|_    SSLv2_RC4_128_EXPORT40_WITH_MD5
|_    SSLv2_RC2_128_CBC_WITH_MD5
|_    SSLv2_DES_64_CBC_WITH_MD5
|_    SSLv2_RC4_128_WITH_MD5
53/tcp    open  domain  ISC BIND 9.4.2
|_dns-nsid:
|_  bind version: 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind  2 (RPC #100000)
|_rpcinfo:
|_  program version  port/proto  service
|_  100000 2 111/tcp  rpcbind
|_  100000 2 111/udp  rpcbind
```

Figure 5: Aggressive Nmap scan displaying operating system and additional service information.

Observation:

The aggressive scan provided additional information about the target operating system and network configuration. This information helps the penetration tester understand the environment and select suitable exploitation techniques.

Step 4: Perform a Comprehensive Scan:

Finally, a comprehensive scan was performed by combining service version detection with aggressive scanning.

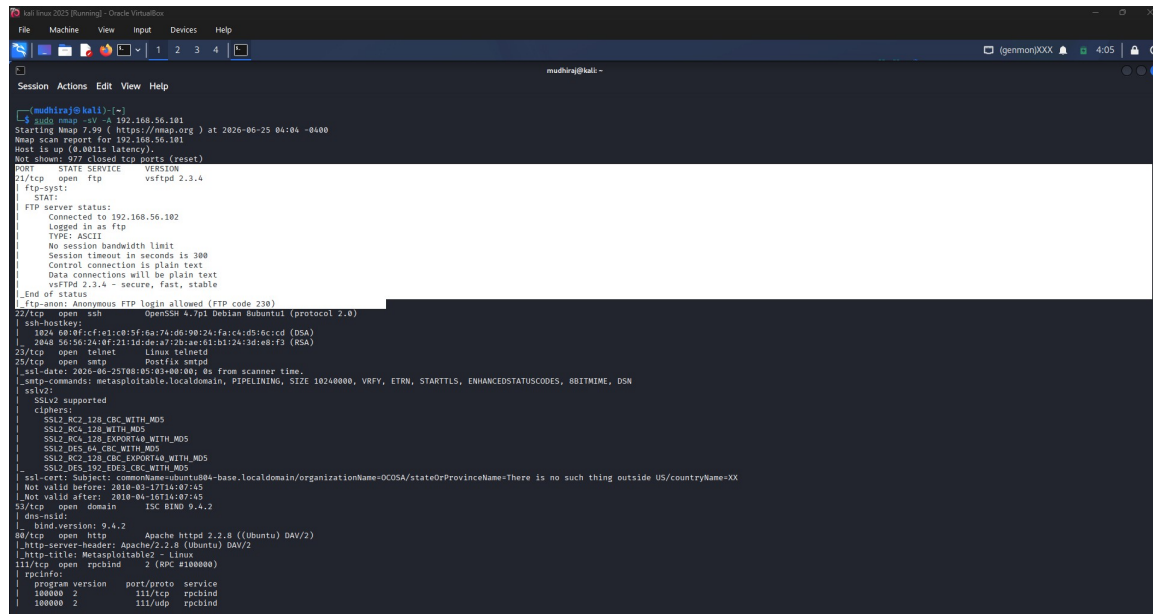
Command Used:

```
sudo nmap -sV -A 192.168.56.101
```

Command Explanation:

- -sV – Detects service versions.
- -A – Performs OS detection, NSE script scanning, and traceroute.
- Combining both options provides a complete overview of the target system.

Screenshot:



```
mudhira@kali:~$ sudo nmap -sV -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-25 04:06 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0012s latency)
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
ftp-syst:
|_ STAT:
|_ FTP server status:
|_   Connected to 192.168.56.102
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
|_ OpenSSH 4.7p1 Debian subuntul (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:8f:cf:a1:c0:5f:6a:7a:d0:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:14:8f:71:1d:0a:27:2b:ae:61:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
|_ Linux telnetd
25/tcp    open  smtp
|_ Postfix satpd
|_ ssl-date: 2026-06-25T04:06:03+00:00; 8s from scanner time.
|_ smtp-command: metasplitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ sslv2:
|_   sslv2 supported
|_   cipher:
|_     SSLv2_RC2_128_CBC_WITH_MD5
|_     SSLv2_RC4_128_WITH_MD5
|_     SSLv2_RC4_128_EXPORT40_WITH_MD5
|_     SSLv2_MD5_64_CBC_WITH_MD5
|_     SSLv2_RC2_128_CBC_EXPORT40_WITH_MD5
|_     SSLv2_MD5_192_128_CBC_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu@base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2018-02-17T14:07:45
|_ Not valid after: 2018-06-10T14:07:45
53/tcp    open  domain
|_ ISC BIND 9.4.2
|_ dnssecid:
|_   bind.version: 9.4.2
80/tcp    open  http
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind
|_ 2 (rpc #100000)
|_ rpcinfo:
|_   program version port/proto service
|_   100000 2 111/tcp rpcbind
|_   100000 2 111/udp rpcbind
```

Figure 6: Comprehensive Nmap scan using -sV -A.

Observation:

The comprehensive scan produced detailed information about the target machine, including open ports, running services, software versions, operating system details, and additional information gathered by Nmap scripts. These results will be used during the exploitation phase.

Scanning Summary:

The scanning phase successfully identified multiple open TCP ports and the services running on the Metasploitable2 target machine. Service version detection revealed outdated software versions that are known to contain publicly documented vulnerabilities. The aggressive scan provided additional operating system and network information. Based on these findings, the target machine was ready for the exploitation phase using Metasploit Framework.